

Gravitationally unstable discs delay the radial drift of dust

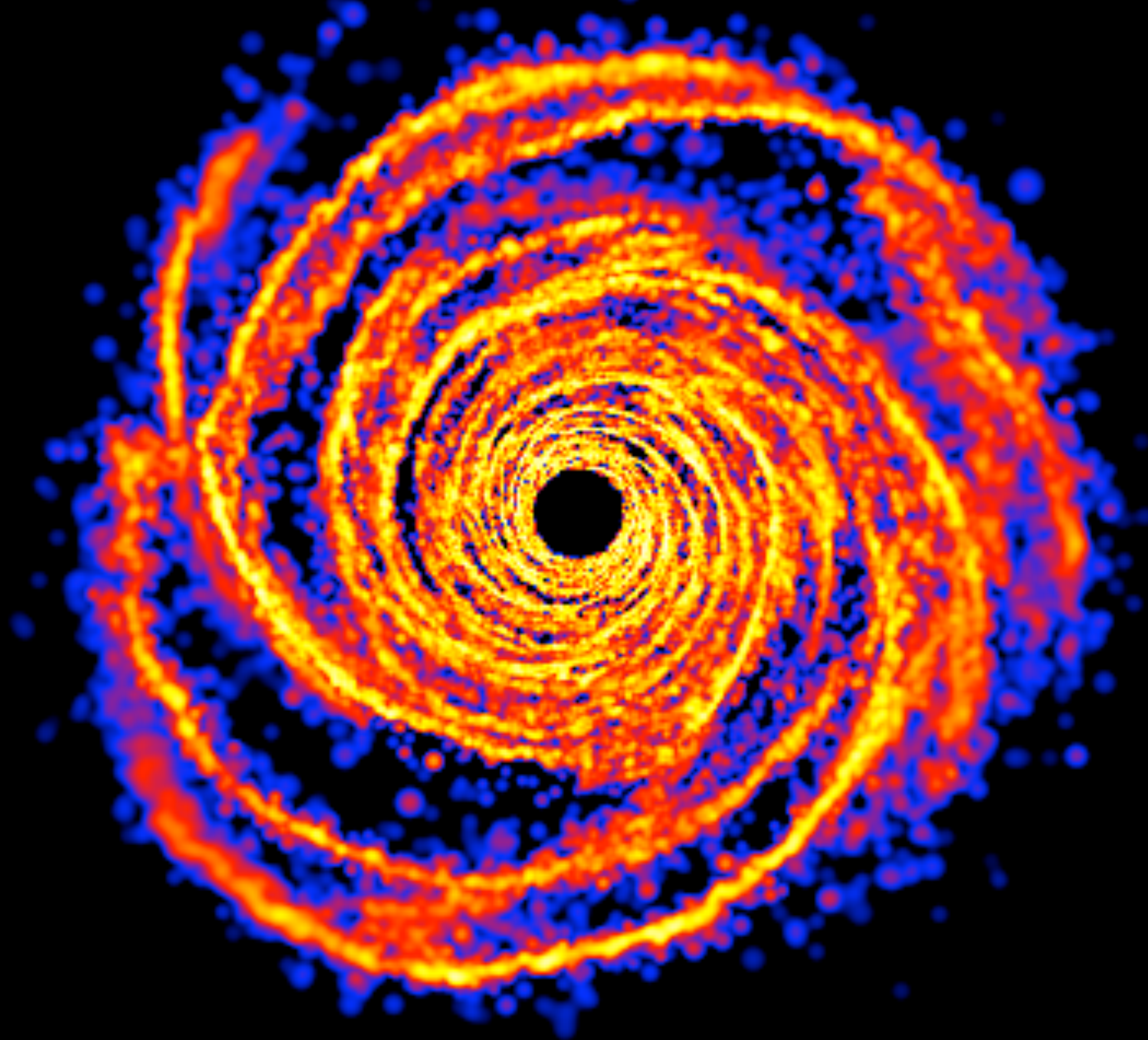
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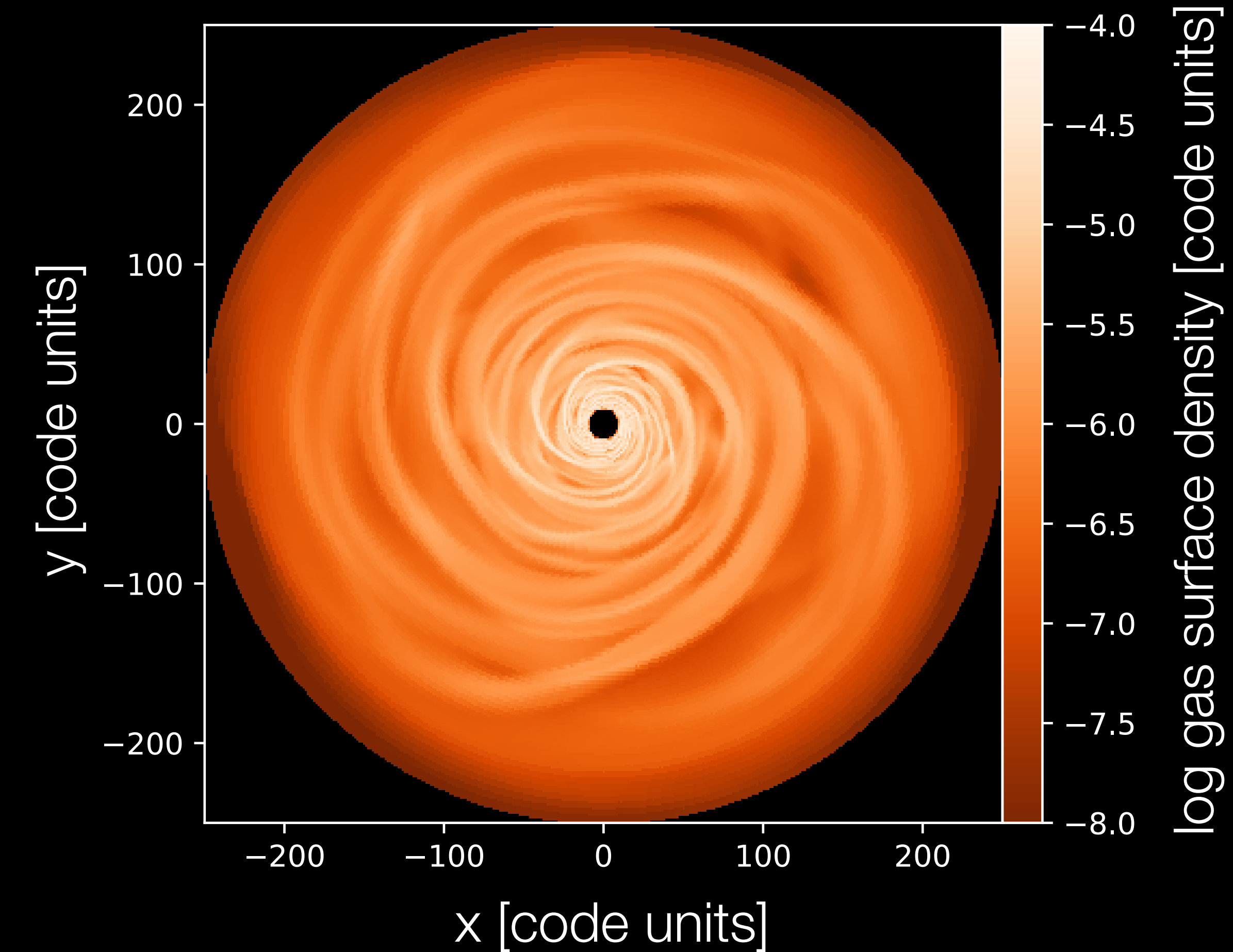
Azimuthal pressure
maxima lead to dust
collecting in the spirals



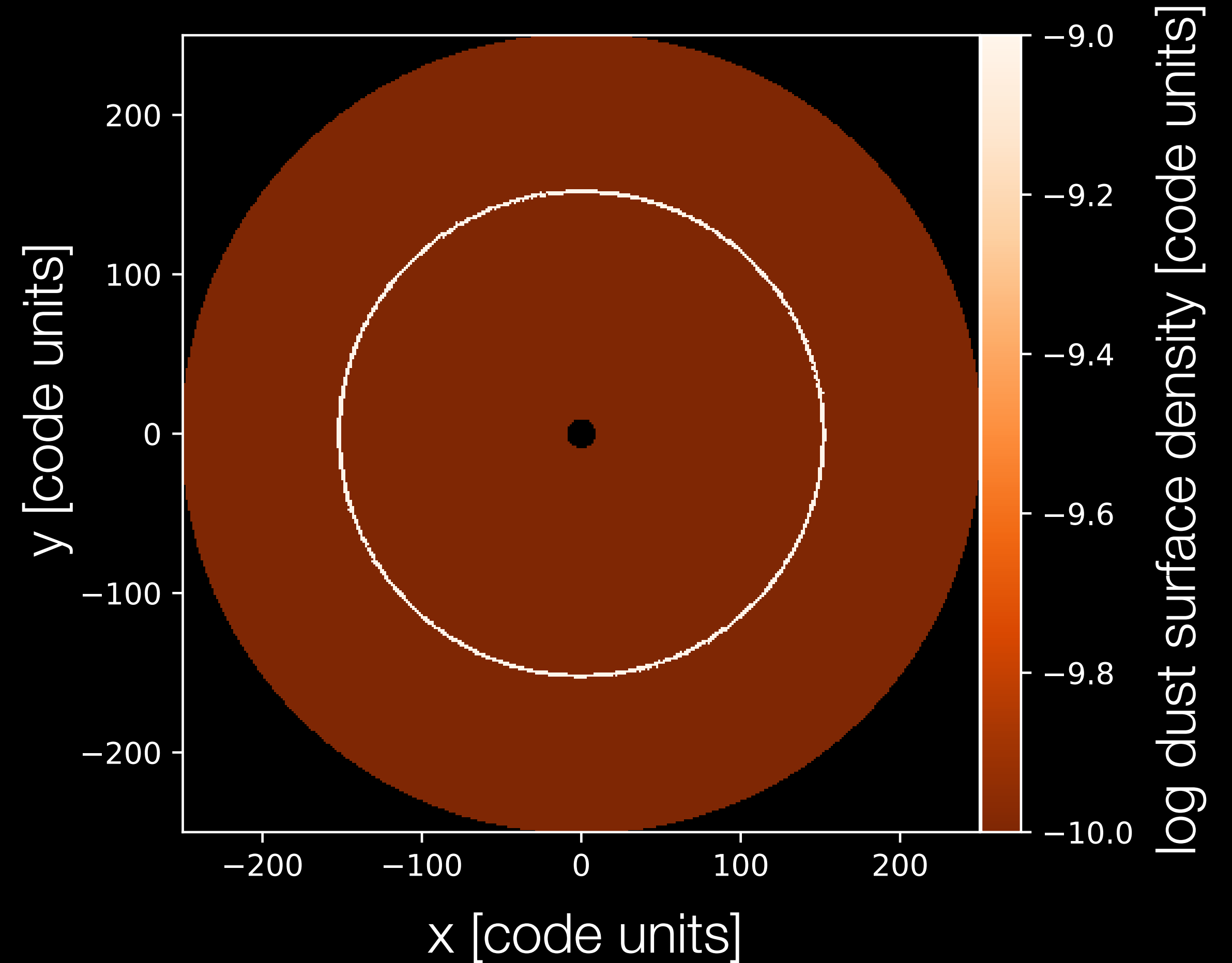
Is the radial drift problem also a
problem in gravitationally unstable
discs?

Controlled radial drift tests

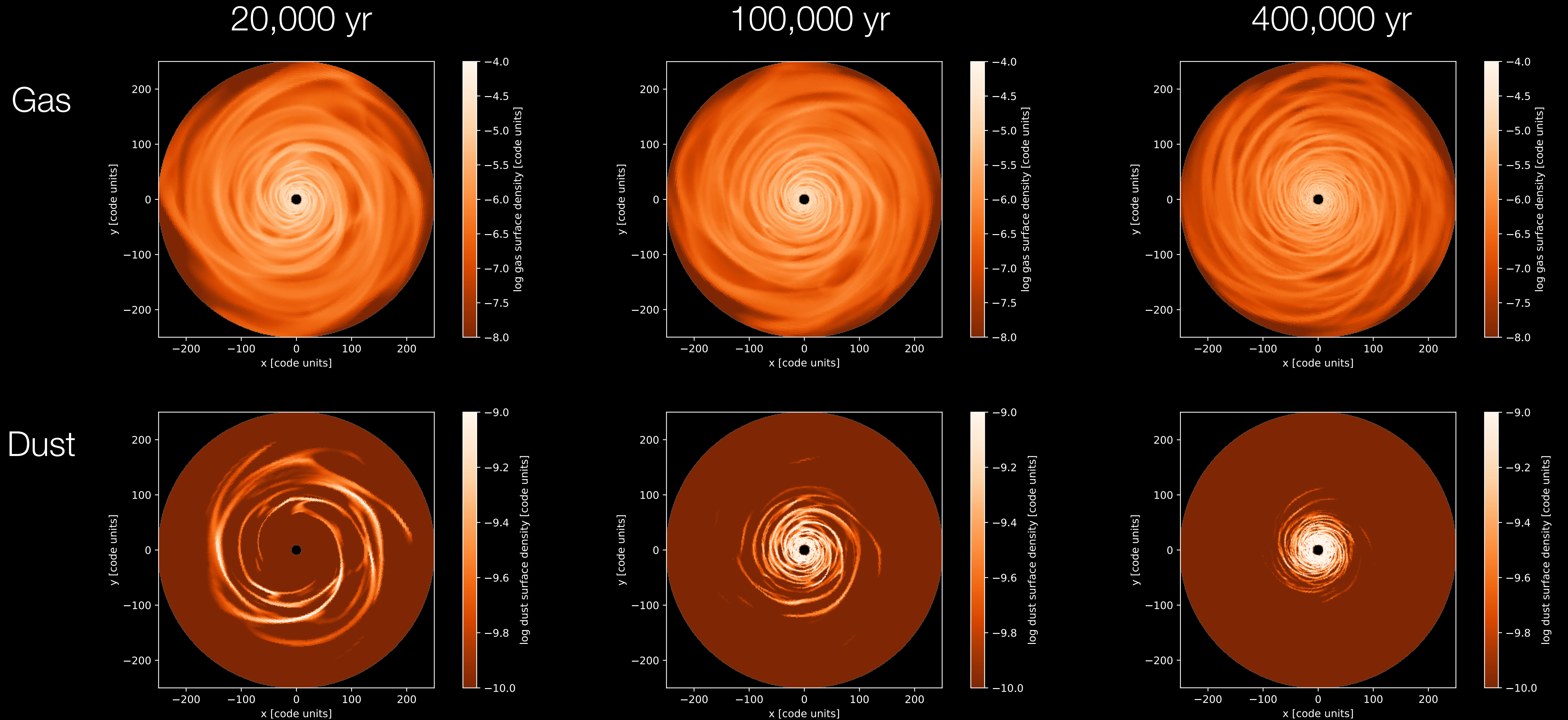
Gas



Dust



Overall we see the dust moving inwards, as expected



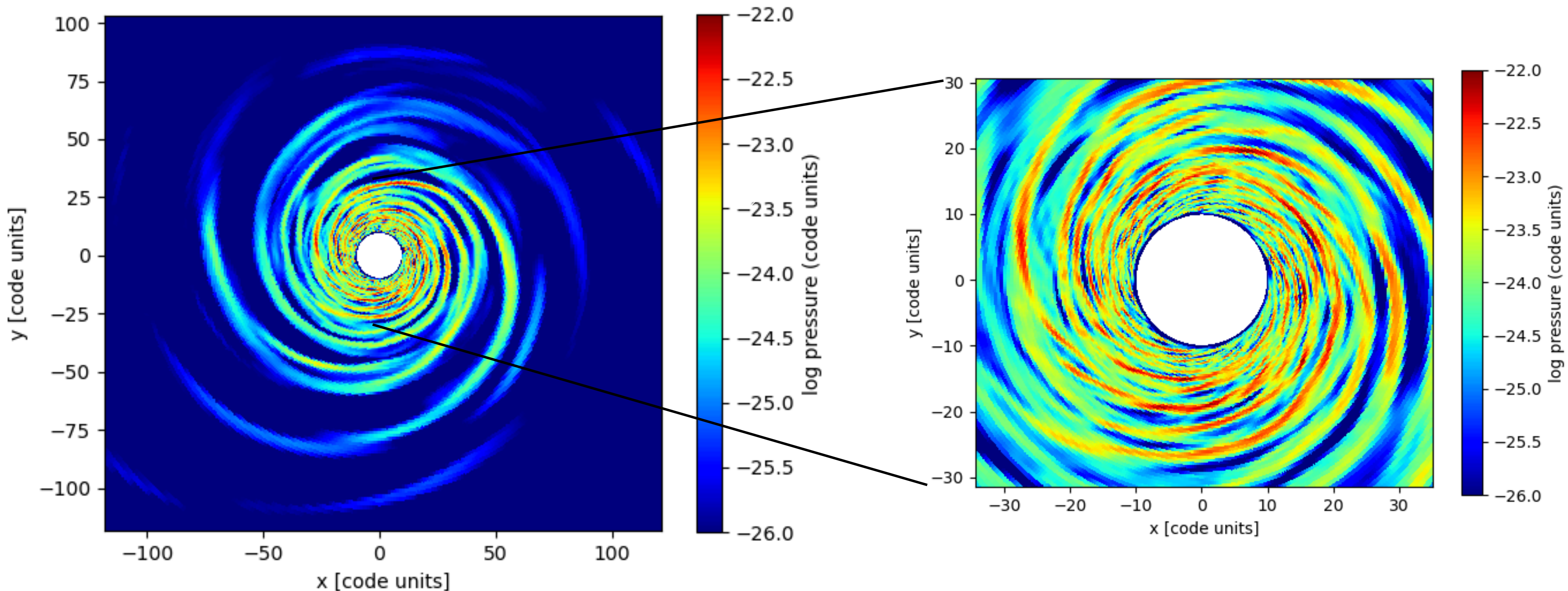
The timescale on which it drifts is longer than expected

Analytical expectation for maximum dust drift
timescale = 580,000 years

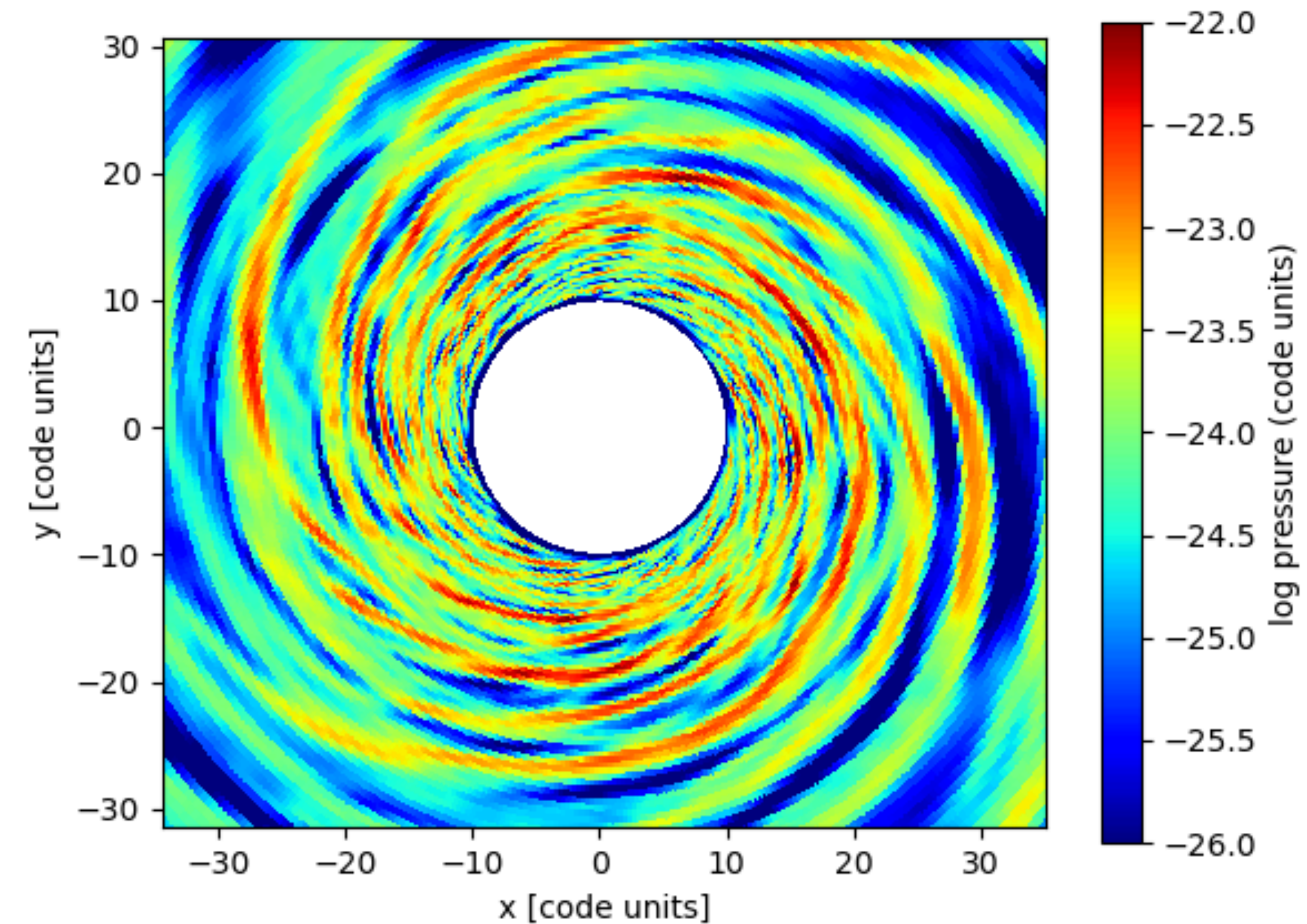
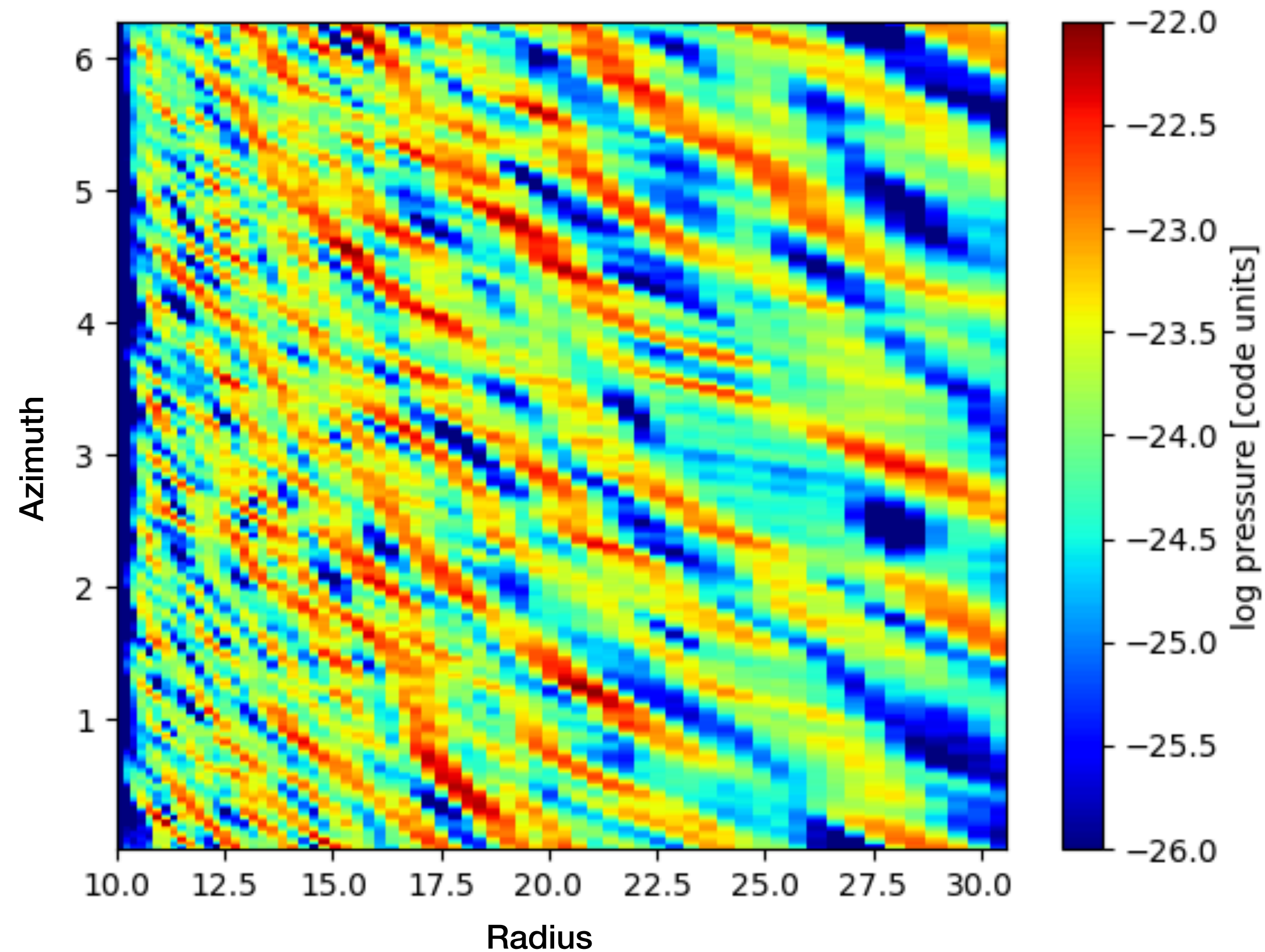
By 800,000 years, only 30% of the dust has been lost

By 1.6 Myr, only 43% of the dust has been lost

There are islands of pressure maxima preventing dust from drifting inwards

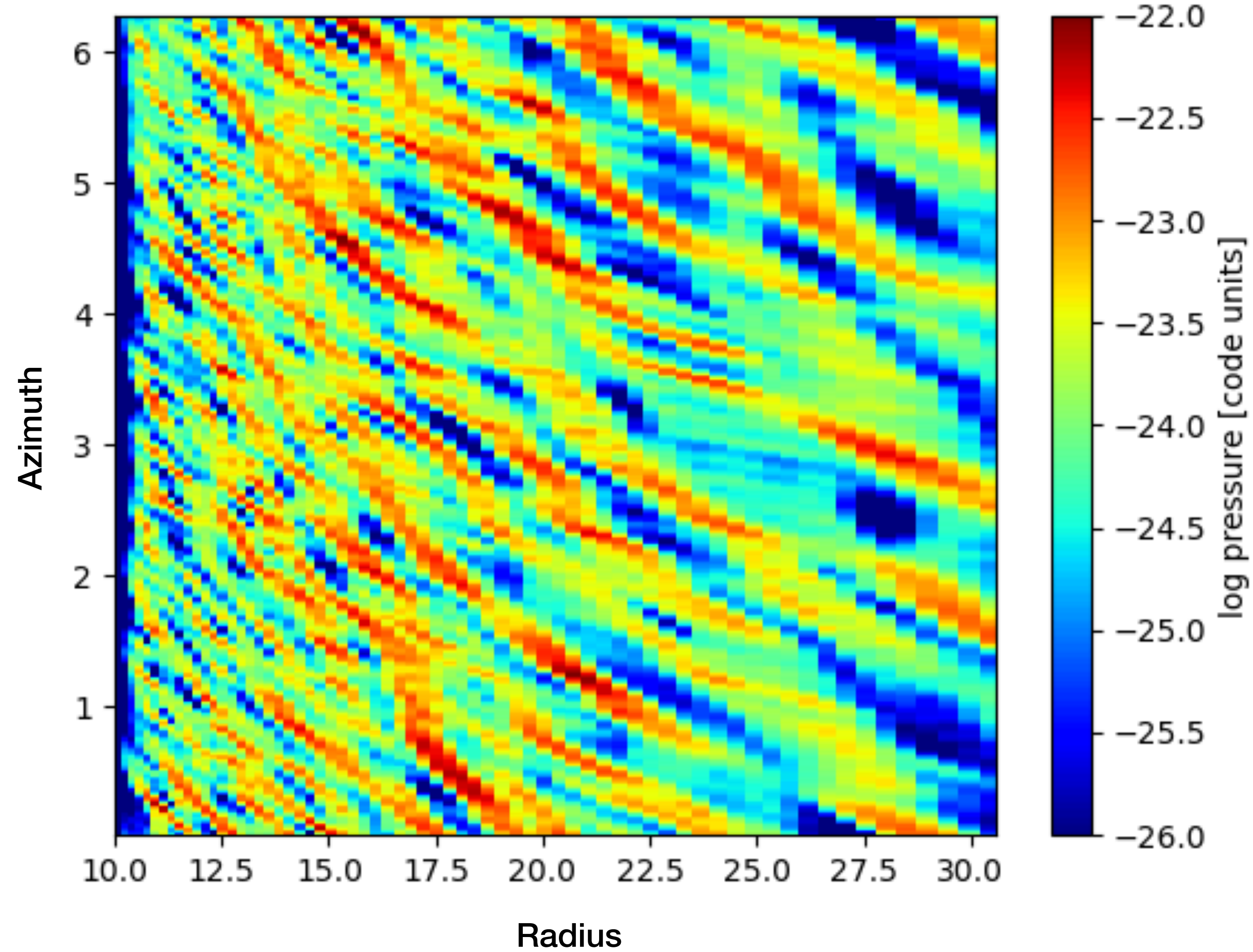


There are islands of pressure maxima preventing dust from drifting inwards



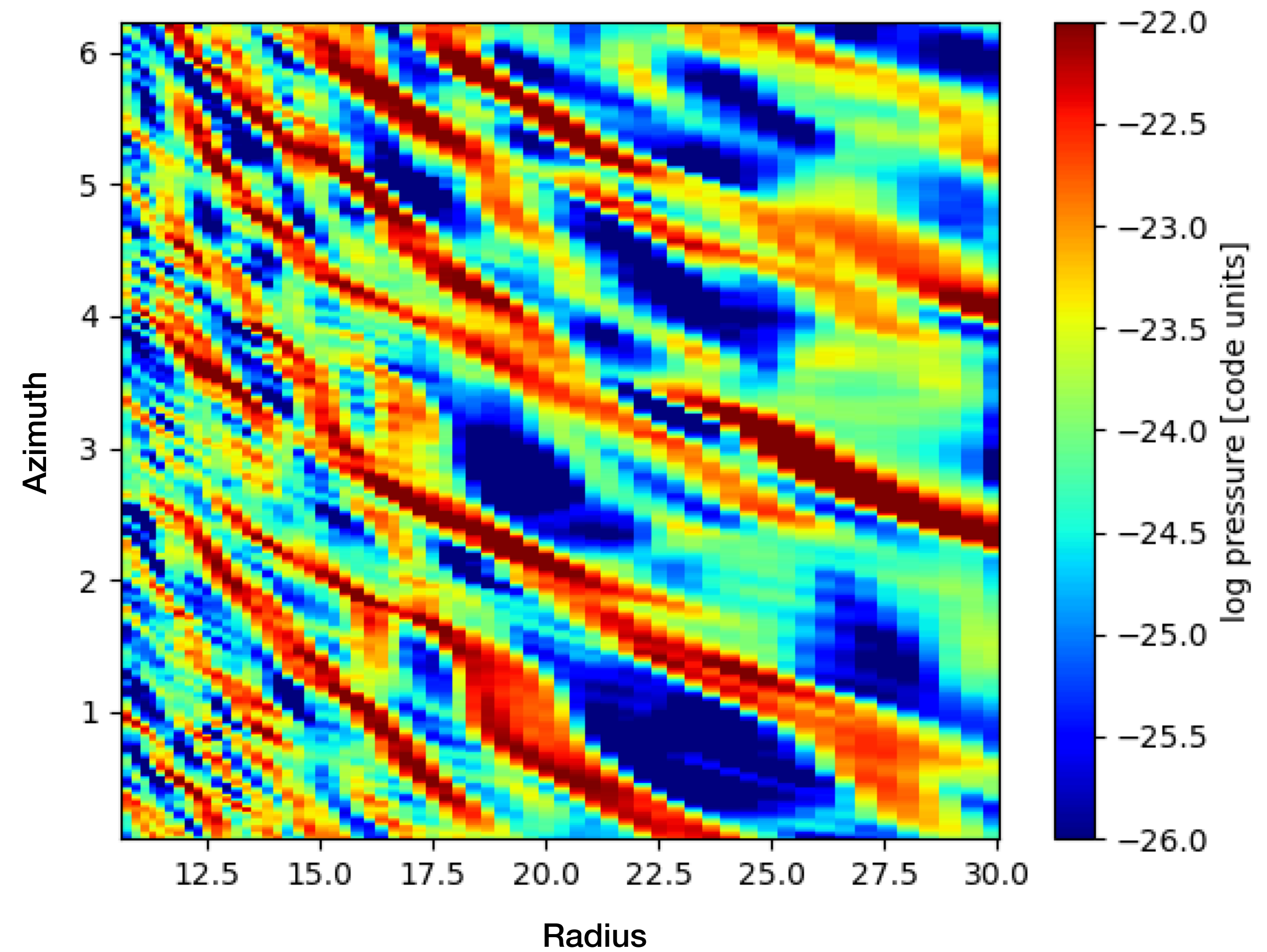
Moderate turbulence leads to stronger pressure islands and slower drift

beta = 10



43% of dust lost by 1.6Myr

beta = 25



20% of dust lost by 1.6Myr

Summary

- ✦ Controlled tests to see how radial drift operates in gravitationally unstable discs
- ✦ Inwards drift is delayed — caused by islands of pressure maxima occurring along the spirals
- ✦ More prominent for moderate amounts of turbulence (higher turbulence washes out the islands)

